Vertex, Axis, Intercepts and Opening Direction
Answer Key

Algebra 1 Parabola Features and Transformations

Quick Answers

1. -4	4. -2, 4, -9	7. -3, 3, -9	10. -1, 5, 9
2. 5, 1	5. -8, -1, 3	8. -2, -6, -2	
3. -9, -5	6. 4, -5	9. 1, 5, -8	

Curriculum

Level: Algebra 1 Parabola Features and Transformations
HSA-SSE.B.3 – *problems 5, 9*
HSF-IF.A–HSF-IF.C – *problems 1, 2, 3, 4, 6, 7, 8, 10*
Difficulty 1–5 of 5 across 20 tagged checks.

Common wrong answers

1. *If they got 32: read the sign of h straight off the parentheses, putting the vertex at x = -3*
3. *If they got 7: used x = b over 2a instead of minus b over 2a, landing on x = 2*
6. *If they got -32: dropped the minus sign on a when computing the axis, landing on x = -3*

1. $y = (x-3)^2 - 4$.

Step 1. This is vertex form $y = a(x-h)^2 + k$ with $a = 1$, $h = 3$, $k = -4$. The subtraction inside the parentheses is what makes h positive.
Step 2. Check by substituting the axis value: $y(3) = (3-3)^2 - 4 = -4$.
Step 3. Vertex $\boxed{(3,\,-4)}$, axis of symmetry $x = 3$, opening upward because $a = 1 > 0$.
Sketch. y-intercept $y(0) = 9 - 4 = 5$; by symmetry the curve also passes through $(6, 5)$. Setting $y = 0$ gives $(x-3)^2 = 4$, so the x-intercepts are 1 and 5 — two units each side of the axis, as symmetry demands.
Common wrong answer: a vertex height of 32 — the sign of h was read straight off the parentheses, putting the vertex at $x = -3$.

2. $y = -(x+2)^2 + 5$.
Step 1. Rewrite the inside as $(x - (-2))$, so $h = -2$ and $k = 5$; the leading coefficient is $a = -1$.
Step 2. $y(-2) = -(0)^2 + 5 = 5$, and because $a < 0$ this is the highest point: the parabola opens downward.
Step 3. Vertex $\boxed{(-2,\ 5)}$, axis $x = -2$, opening downward.
Step 4. y-intercept: $y(0) = -(0+2)^2 + 5 = -4 + 5 = \boxed{1}$
Sketch. Symmetric partner of the y-intercept is $(-4, 1)$. Setting $y = 0$ gives $(x+2)^2 = 5$, so the x-intercepts are about -4.2 and 0.2.

3. $y = x^2 + 4x - 5$.
Step 1. Standard form with $a = 1$, $b = 4$, $c = -5$. Axis of symmetry: $x = -\dfrac{b}{2a} = -\dfrac{4}{2} = -2$.
Step 2. Substitute to get the vertex height: $y(-2) = 4 - 8 - 5 = -9$.
Step 3. Vertex $\boxed{(-2,\ -9)}$, axis $x = -2$, opening upward.
Step 4. The y-intercept is c: $y(0) = \boxed{-5}$
Sketch. Factoring $(x+5)(x-1)$ gives x-intercepts -5 and 1, and they straddle $x = -2$ as they must.
Common wrong answer: 7 — $x = \frac{b}{2a}$ was used instead of $-\frac{b}{2a}$, putting the axis at $x = 2$.

4. $y = x^2 - 2x - 8$.
Step 1. The x-intercepts are where $y = 0$, so factor: two numbers with product -8 and sum -2 are -4 and 2.
Step 2. $x^2 - 2x - 8 = (x-4)(x+2)$, so $x - 4 = 0$ or $x + 2 = 0$.
Step 3. x-intercepts $\boxed{x = -2 \text{ and } x = 4}$
Step 4. The axis of symmetry is midway between them: $x = \frac{-2+4}{2} = 1$, and $y(1) = 1 - 2 - 8 = -9$.
Step 5. Vertex $\boxed{(1,\ -9)}$, opening upward.
Sketch. y-intercept -8, with symmetric partner $(2, -8)$.

5. $y = 2(x-1)^2 - 8$.
Step 1. Vertex form with $h = 1$, $k = -8$, $a = 2$, so $y(1) = \boxed{-8}$ and the vertex is $(1, -8)$, a minimum.
Step 2. For the x-intercepts set $y = 0$: $2(x-1)^2 = 8$, so $(x-1)^2 = 4$ and $x - 1 = \pm 2$.
Step 3. x-intercepts $\boxed{x = -1 \text{ and } x = 3}$
Step 4. Since $|a| = 2 > 1$, the parabola is *narrower* than $y = x^2$: one unit from the axis it has already risen 2 instead of 1.
Sketch. Axis $x = 1$, y-intercept $y(0) = -6$.

6. $y = -x^2 + 6x - 5$.

Step 1. Here $a = -1$, $b = 6$, $c = -5$. Axis of symmetry: $x = -\dfrac{6}{2(-1)} = 3$. Keeping the minus sign on a is the whole difficulty.
Step 2. $y(3) = -9 + 18 - 5 = 4$. Because $a < 0$ this point is a *maximum* and the parabola opens downward.
Step 3. Vertex $\boxed{(3,\ 4)}$, a maximum, axis $x = 3$.
Step 4. y-intercept: $y(0) = \boxed{-5}$
Sketch. $-x^2 + 6x - 5 = -(x-1)(x-5)$, so the x-intercepts are 1 and 5, symmetric about $x = 3$.
Common wrong answer: -32 — the minus sign on a was dropped in $-\frac{b}{2a}$, putting the axis at $x = -3$.

7. $y = x^2 - 9$.
Step 1. Set $y = 0$: $x^2 = 9$. This is a difference of squares, $(x-3)(x+3) = 0$.
Step 2. x-intercepts $\boxed{x = -3 \text{ and } x = 3}$
Step 3. There is no x term, so $b = 0$ and the axis of symmetry is $x = -\frac{0}{2} = 0$ — the y-axis itself. That is exactly why the two intercepts are mirror images of each other.
Step 4. $y(0) = -9$, so the vertex is $\boxed{(0,\ -9)}$, a minimum (the parabola opens upward).
Sketch. Symmetric points $(\pm 2, -5)$ help fix the width.

8. $y = \frac{1}{2}(x+4)^2 - 2$.
Step 1. Vertex form with $h = -4$, $k = -2$ and $a = \frac{1}{2}$, so $y(-4) = \boxed{-2}$ and the vertex is $(-4, -2)$.
Step 2. Set $y = 0$: $\frac{1}{2}(x+4)^2 = 2$, so $(x+4)^2 = 4$ and $x + 4 = \pm 2$.
Step 3. x-intercepts $\boxed{x = -6 \text{ and } x = -2}$
Step 4. $a = \frac{1}{2} > 0$, so it opens upward, and since $|a| < 1$ it is *wider* than $y = x^2$: one unit from the axis it rises only $\frac{1}{2}$.
Sketch. Axis $x = -4$; y-intercept $y(0) = 8 - 2 = 6$.

9. $y = 2x^2 - 12x + 10$.
Step 1. Every coefficient is even, so factor the leading coefficient out first: $2(x^2 - 6x + 5) = 2(x-1)(x-5)$.
Step 2. A product is zero when a factor is zero, and the 2 can never be zero, so $x = 1$ or $x = 5$.
Step 3. x-intercepts $\boxed{x = 1 \text{ and } x = 5}$
Step 4. Midpoint of the intercepts: $x = \frac{1+5}{2} = 3$. Check against the formula: $-\dfrac{-12}{2(2)} = 3$ ✓. Then $y(3) = 18 - 36 + 10 = -8$.
Step 5. Vertex $\boxed{(3,\ -8)}$, opening upward since $a = 2 > 0$.
Sketch. y-intercept 10, symmetric partner $(6, 10)$.

10. **Intercepts -1 and 5, through $(2, 9)$.** *The sketch and the explanation are open work, flagged for manual review; the model answer below is what full credit contains.*

Step 1. Knowing both x-intercepts fixes the factors, so the equation must look like $y = a(x + 1)(x - 5)$. Only a is unknown, and one more point is exactly enough to find it.

Step 2. Substitute $(2, 9)$: $9 = a(2 + 1)(2 - 5) = a(3)(-3) = -9a$, so $a = -1$ and $y = -(x + 1)(x - 5)$.

Step 3. Multiply out: $-(x^2 - 4x - 5) = -x^2 + 4x + 5$. Its x-intercepts are still $\boxed{x = -1 \text{ and } x = 5}$

Step 4. The axis is midway between the intercepts, at $x = \frac{-1+5}{2} = 2$, and the vertex height is $y(2) = -4 + 8 + 5 = \boxed{9}$

Step 5 (the sketch). Vertex $(2, 9)$, axis $x = 2$, opening downward because $a = -1 < 0$; y-intercept 5.

Full credit: the equation $y = -(x + 1)(x - 5) = -x^2 + 4x + 5$, a labelled sketch, and a sentence saying that the point $(2, 9)$ sits *above* the x-axis between the two intercepts, which can only happen if the parabola arches over — forcing a to be negative.